

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS
ON EXPERIENCE (IOHE)
ON
SITEKRAFT – AI POWERED WEBSITE BUILDER**



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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Sitekraft – AI Powered Website Builder**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Monika Aggarwal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 15 May 2026

Koustubh Kukreti

(2210990517)

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

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We are thankful to the Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, for providing us with the opportunity to undertake this Industry Oriented Hands on Experience (IOHE) project.

We also acknowledge the open-source communities and documentation resources related to React, FastAPI, Vercel, and Google Gemini AI that contributed significantly to the successful completion of the project.

ABSTRACT

Sitekraft is an autonomous AI-powered website builder capable of generating production-ready React websites from simple natural language business descriptions. The platform integrates Google Gemini AI, FastAPI, React, Vite, contextual image APIs, AI copywriting systems, and automated cloud deployment to create professional websites without requiring coding expertise from users.

The system follows a modular client-server architecture where the frontend collects structured business information through an onboarding workflow, while the backend orchestrates AI code generation, content writing, image integration, and deployment automation. The generated websites include responsive layouts, modern UI sections, professional marketing content, downloadable source code, and live cloud-hosted deployment links.

Unlike conventional template-based builders, Sitekraft generates real React JSX component structures with customizable themes, responsive layouts, and business-specific sections. The project demonstrates practical application of large language models in autonomous software engineering workflows and reduces the technical barriers associated with modern web development.

Introduction

Website development has become essential for businesses in the digital economy. However, traditional website creation requires knowledge of frontend development, backend systems, responsive design, deployment workflows, and cloud infrastructure. Small businesses and non-technical users often struggle with these technical requirements.

Sitekraft addresses this problem by providing an AI-powered autonomous website generation platform. Users simply provide business information such as category, services, style preference, and color theme, and the system generates a complete deployable React website automatically.

Problem Statement:

Traditional website development is expensive, time-consuming, and technically complex. Existing drag-and-drop builders provide limited flexibility and rely heavily on static templates. There is a need for an intelligent AI-based system capable of generating unique production-ready websites autonomously.

Objectives:

- Generate React websites automatically using AI
- Provide one-click deployment to cloud platforms.
- Support multiple business categories dynamically
- Integrate AI-generated professional copywriting
- Reduce technical barriers for non-technical users

Methodology

The project followed a modular agile development methodology divided into requirement analysis, architecture planning, implementation, testing, and deployment.

Requirement Analysis

Requirements were gathered for onboarding workflows, AI code generation, deployment automation, image integration, and responsive design support.

System Architecture

The system follows a three-layer architecture:

- Frontend Layer – React + Vite UI
- Backend Layer – FastAPI orchestration services
- External Services – Gemini AI, Pexels API, Pixabay API, Vercel API

AI Workflow

- User enters business information
- Backend processes structured business context
- Gemini AI generates React JSX components
- AI copywriter generates marketing content
- Image APIs fetch contextual images
- Vercel API deploys generated website automatically

Development Process

The frontend onboarding flow was developed first, followed by backend AI orchestration and deployment modules. Each module was tested independently before integration.

Tools and Technologies

Frontend Technologies:

- React 19
- Vite
- JavaScript
- CSS
- Axios
- JSZip

Backend Technologies:

- Python 3.12
- FastAPI
- Uvicorn
- Pydantic

AI & APIs:

- Google Gemini 2.5 Flash
- Groq API
- OpenRouter
- Pexels API
- Pixabay API

Deployment & Cloud:

- Vercel
- Render
- Supabase

Development Tools:

- Git & GitHub
- VS Code
- Postman

Implementation**Frontend Implementation**

The frontend onboarding system was implemented using React functional components with state-driven navigation. Users complete multiple onboarding steps including business category selection, business name, services, theme selection, and image uploads.

Backend Implementation

The backend FastAPI server manages AI orchestration, API routing, deployment automation, and code generation workflows. The backend processes structured business information and coordinates multiple AI modules.

AI Code Generation

Google Gemini AI generates production-ready React JSX components including:

- Navbar
- Hero Section
- Services Section
- About Section
- Contact Section
- Footer

Image Integration

The system integrates Pexels and Pixabay APIs to fetch contextual business images automatically.

Deployment System

The generated Vite project structure is packaged and deployed automatically using the Vercel Deployment API. The system returns a publicly accessible deployment URL to the user.

Results

The project successfully demonstrated autonomous AI-based website generation and deployment.

Key Results:

- Average AI code generation time: 15–25 seconds
- Average deployment time: Under 70 seconds
- Responsive website generation across business categories
- Successful deployment using Vercel cloud infrastructure
- Dynamic AI-generated marketing content
- High accessibility and performance scores

Business Categories Tested:

- Restaurants
- Salons
- Coaching Centers
- Online Stores
- Real Estate Agencies

Key Features Delivered:

- AI-powered React code generation
- Professional AI copywriting
- Responsive UI layouts
- Context-aware image integration
- ZIP export support
- One-click deployment

Conclusion and Future Scope

Sitekraft successfully demonstrates how large language models and AI orchestration systems can automate complete website development workflows. The platform combines AI-generated code, automated deployment, contextual content generation, and responsive frontend systems into a single integrated architecture. The project significantly reduces technical barriers for small businesses and demonstrates practical use of AI in software engineering automation.

- Multi-page website support
- Custom domain integration
- SEO optimization
- Real-time AI streaming
- Post-deployment editing
- Mobile application support
- Analytics integration

REFERENCES

- Google Gemini API Documentation
- React Documentation
- FastAPI Documentation
- Vercel Deployment Documentation
- <https://ai-agent-gold-rho.vercel.app>

APPENDICES

GitHub Repository:

<https://github.com/koustubhkukreti/Sitekraft-AI-Powered-Website-Builder-2210990517-2210990857>

Live Deployment:

<https://ai-agent-gold-rho.vercel.app>

Copyright Receipt

Online Services

- E-Filing of Copyright Application
 - Registration of copyright (Form-XIV)
 - Change in Particulars (Form-XV)
- E-filing of Application for Copyright Society
 - Registration/renewal of Copyright Society (Form VIII & IX)
 - Registration/renewal of Performers' Society (Form XI & XII)
- Status of the Application
- Deficiency in Copyright Applications
- Received Application
- Work Awaited Application
- Application for Hearing
- Details of Processing Fee
- Check List
- Workflow
- FAQ
- Document Upload Manual
- Cancellation/Rectification Petition Notices and Stay Orders by Courts
- Expunged ROC
- Upload Work
- Notice

Copyright Receipt

भारत सरकार / GOVERNMENT OF INDIA
कोपीराइट कार्यालय / Copyright Office
बौद्धिक संपदा भवन, प्लॉट नंबर 32, सेक्टर 14, द्वारका, नई दिल्ली-110078 फोन: 011-28032496
Intellectual Property Bhawan, Plot No. 32, Sector 14, Dwarka, New Delhi 110078 Phone: 011-28032496
रसीद / Receipt

To: Snehita Chaudhary
2902 near panaji plant-140401
(M)-7573235351 E-mail-chaudharysneha53@gmail.com

RECEIPT NO : 240456
FILING DATE : 29/04/2025
BRANCH : Delhi
USER : sneha_chaudhary

PAGE No : 1

S.No	Form	Diary No.	Request No	Title	Amount (Rupees)
1	Form XIV	ISW-202302025-CO	272536	SITEKRAFT-AI-POWERED WEBSITE BUILDER	500
Amount in Words					Rupees Five Hundreds

PAYMENT MODE	Transaction id	CIN
Online	C-0000311568	2804250000387

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Useful Links

- Department For Promotion of Industry & Internal Trade
- World Intellectual Property Organization
- Intellectual Property India

SHe-Box

Screenshots

Chrome File Edit View History Bookmarks Profiles Tab Window Help

100% 85% Fri 15 May 2:33 AM

Sitekraft

ai-agent-gold-rho.vercel.app

Sitekraft

Step 1 of 8

What type of business is this?

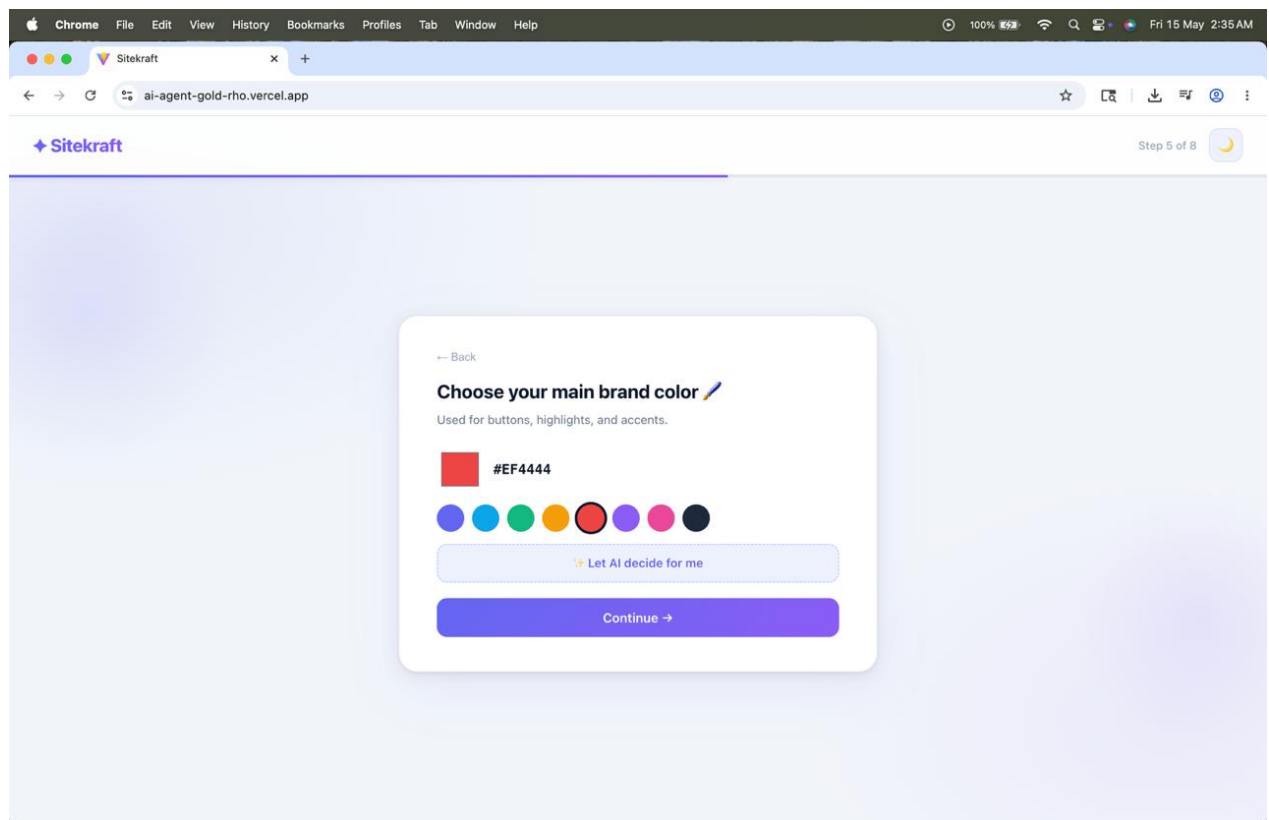
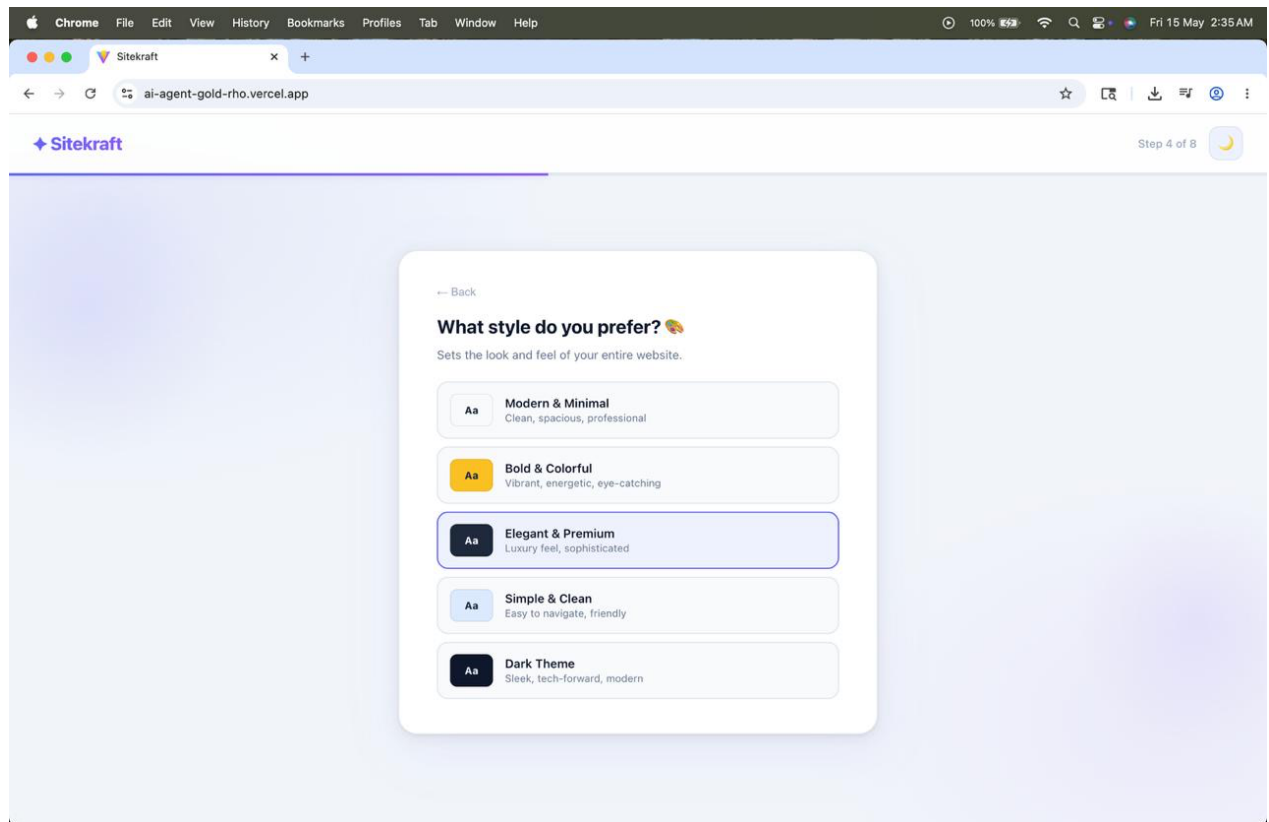
Pick the one that fits best — you can fine-tune after.

- Restaurant / Café
- Clothing Store
- Salon / Spa
- Gym / Fitness
- Real Estate
- Coaching / Classes
- Online Store
- Other Business

Description of the App

The screenshot shows a web browser window with the Sitekraft app running. The browser's address bar displays 'ai-agent-gold-rho.vercel.app'. The app's header includes the Sitekraft logo and 'Step 2 of 8' with a moon icon. The main content area features a white card with a 'Back' link. The card contains the text 'What's your business called?' followed by a smiley face emoji and a subtext 'Don't worry — you can always change this later.' Below this is a text input field containing 'Koala Coffee' and a blue 'Continue →' button.

The screenshot shows the same web browser window with the Sitekraft app at 'Step 3 of 8'. The main content area features a white card with a 'Back' link. The card contains the text 'Where is your business located?' followed by a location pin icon and a subtext 'Just city and country is enough.' Below this is a text input field containing 'Chandigarh, India' and a blue 'Continue →' button.



Sitekraft

Koala Coffee

+

ai-agent-gold-rho.vercel.app

Coursea Certifica...

Step 6 of 8

← Back

Tell us about your products 📷

Add the products you want to showcase on your site.

PRODUCT 1

Product name * — e.g. Dal Makhni

Price — optional e.g. ₹240

Short description — optional e.g. Slow-cooked black lentils in rich buttery tomato gravy

Category — optional e.g. Main Course / Starters

Subcategory — optional e.g. North Indian / South Indian

Add photo — optional

+ Add Another Product

Continue →

Sitekraft

Koala Coffee

+

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Coursea Certifica...

Step 7 of 8

← Back

Do you have business photos to upload? 📷

Great photos make websites shine — but we can also find them for you.

 **Yes, I'll upload my own photos**
Logo, storefront, team

 **Use AI-generated + stock images**
We'll find the best visuals automatically

Result

